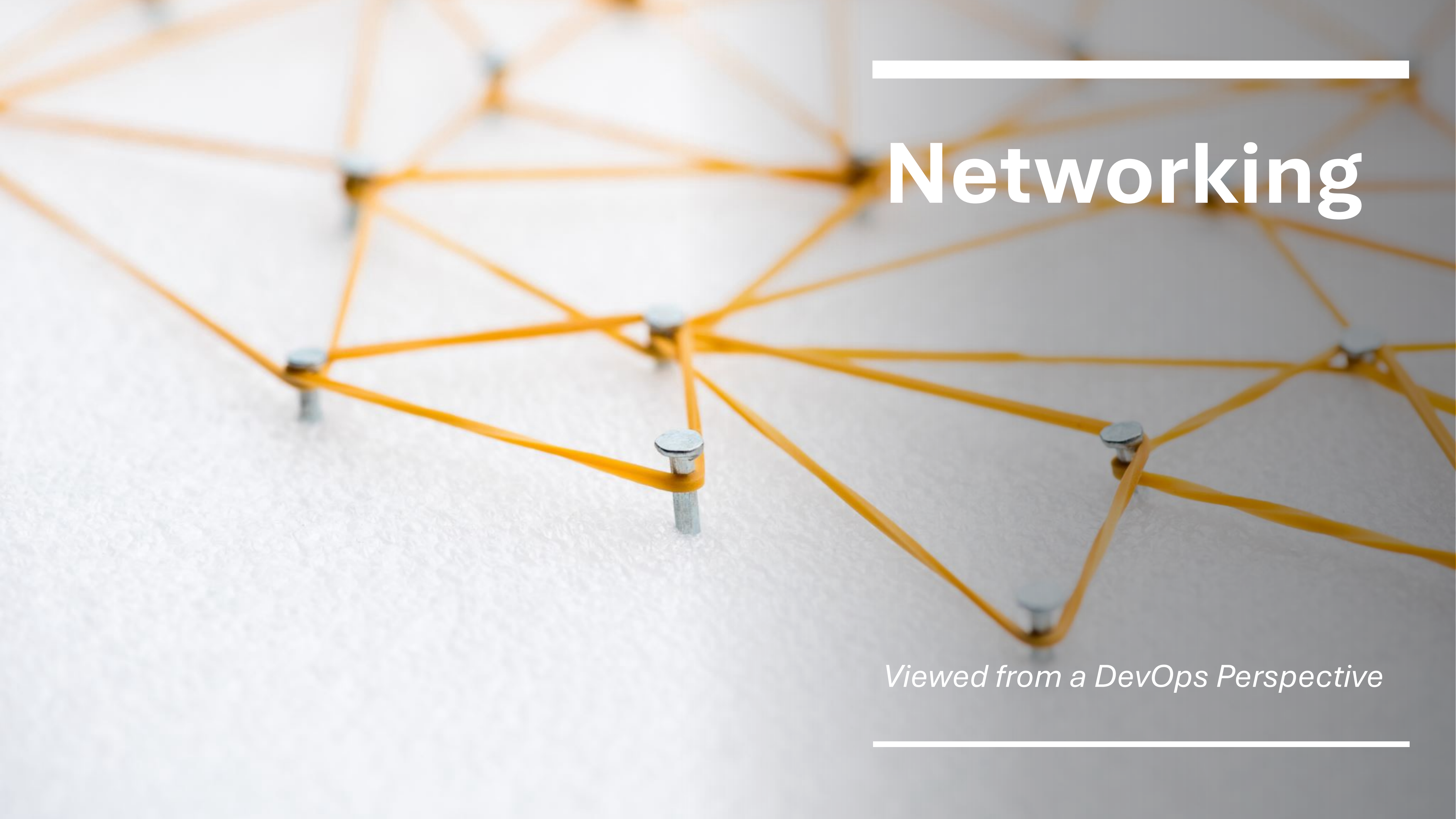


Networking

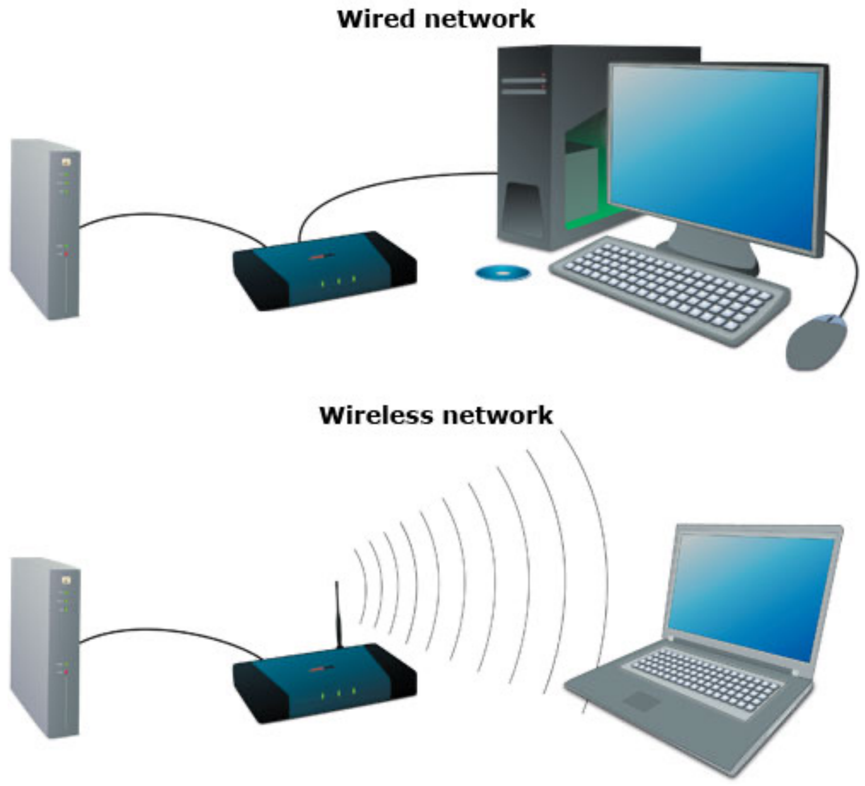
Viewed from a DevOps Perspective

Contents

1. Components
2. OSI Model
3. Classification
4. Devices
5. Home Network
6. IP Addresses
7. Protocols
8. DNS & DHCP
9. Network Commands

Computer Networking

Communication between two or more network Interfaces.



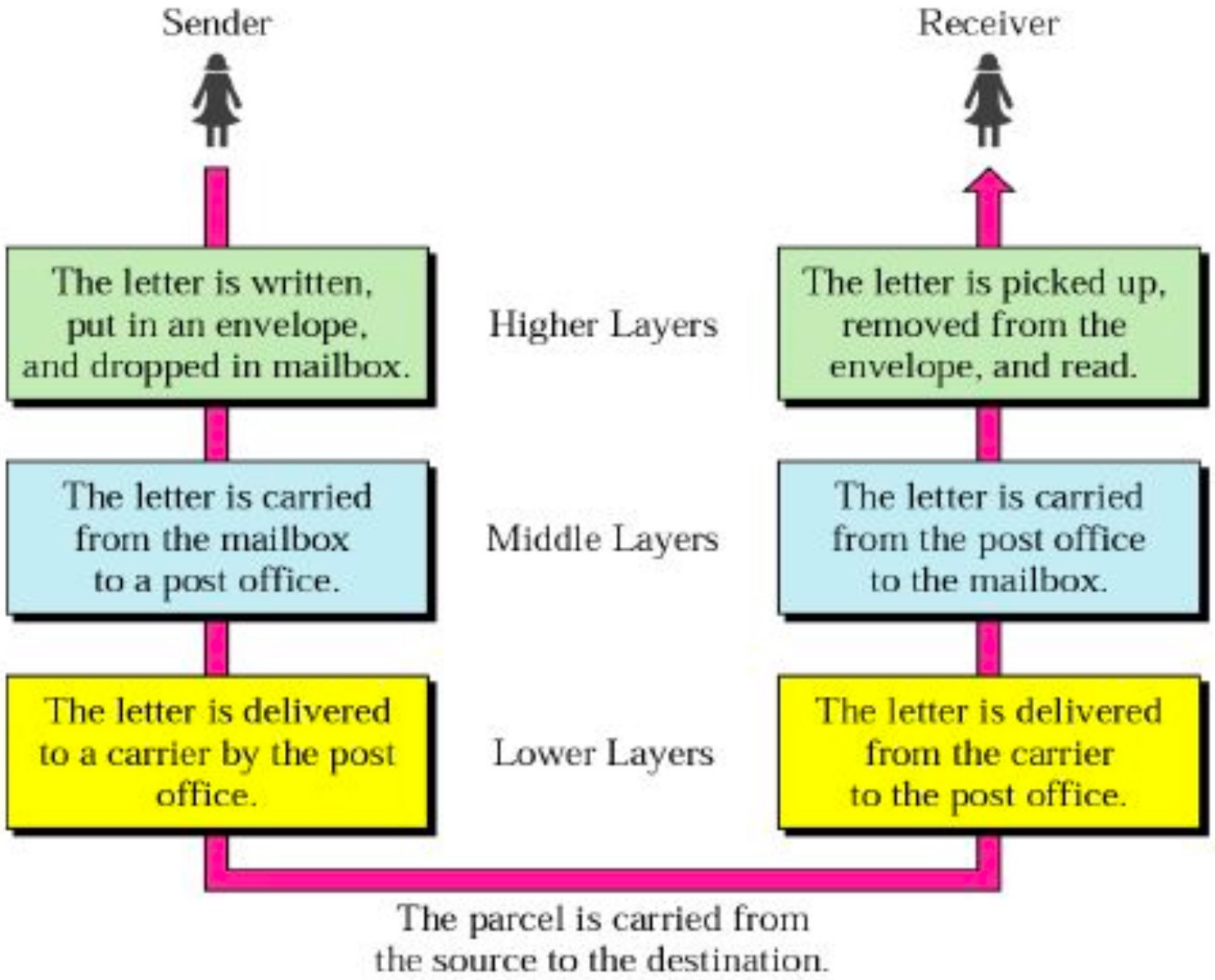
OSI Model

● People around the world uses computer network to communicate with each other.
● For worldwide data communication, systems must be developed which are compatible to communicate with each other.
● There should be standard communication methods & devices.
● ISO (International Organization of Standardization) has developed this standard.
● This communication model is called as Open System Interconnection (OSI).
● ISO-OSI model is a seven layer architecture developed in 1984.

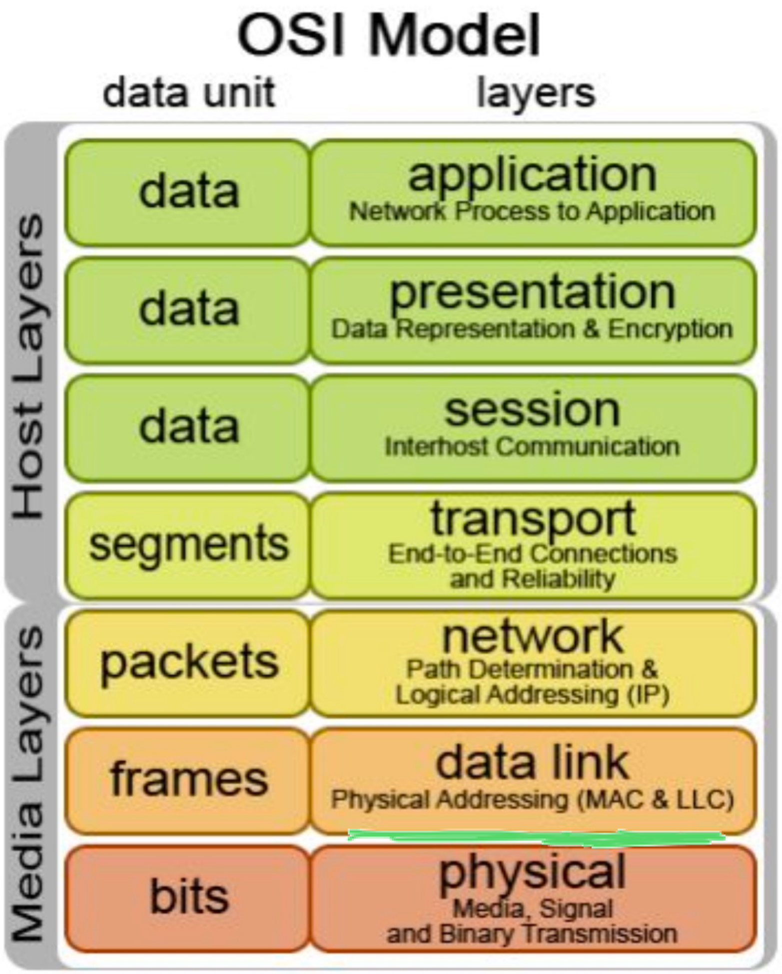
OSI Model

- The basic elements of a layered model are
  - services
  - protocols
  - and interfaces.
- A service is a set of actions that a layer offers to another (higher) layer.
- A Protocol is a set of rules that a layer uses to exchange information.
- A Interface is communication between the layers.

Sending - Receiving Letters



OSI Model

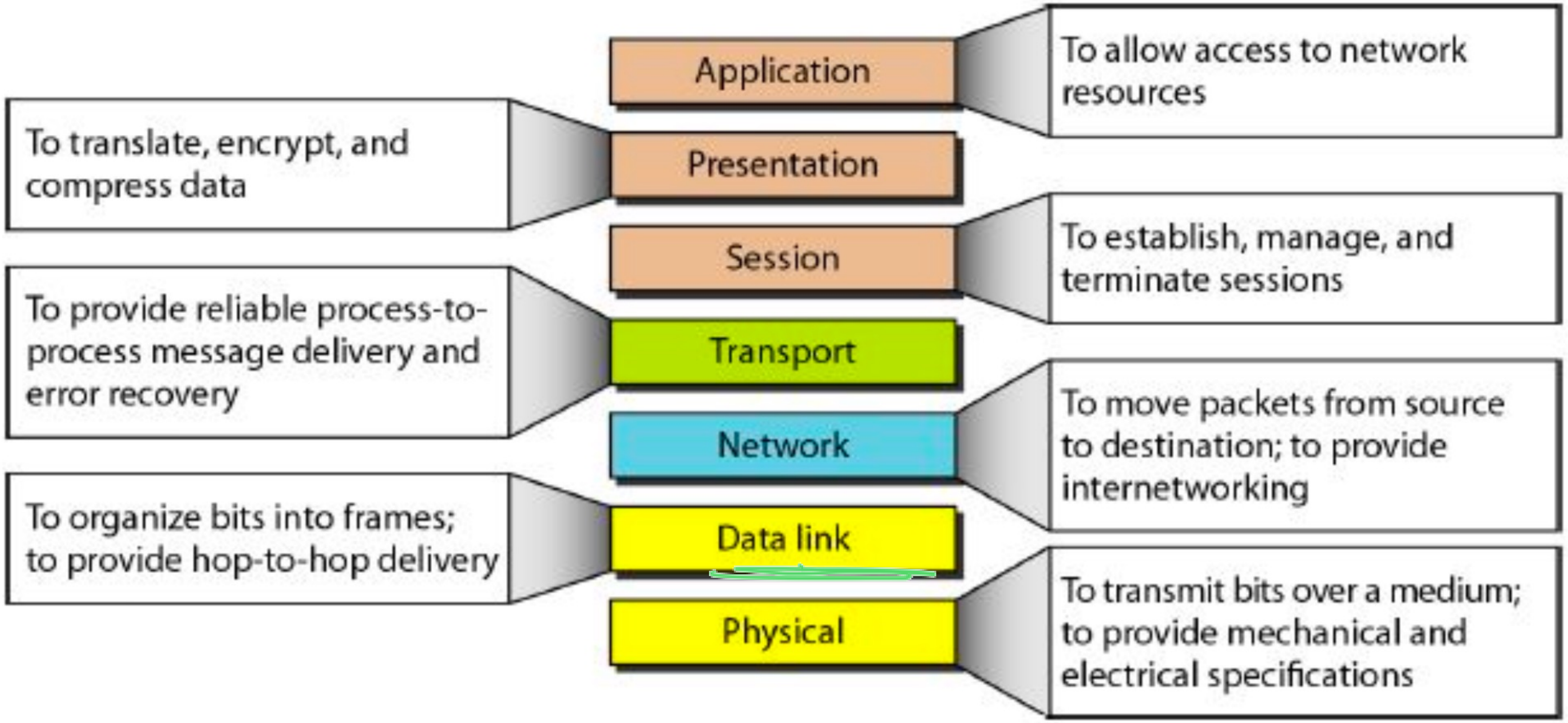


OSI Model
data unit
layers
Host Layers
data
application
Network Process to Application
data
presentation
Data Representation & Encryption
data
session
Interhost Communication
segments
transport
End-to-End Connections and Reliability
Media Layers
packets
network
Path Determination & Logical Addressing (IP)
frames
data link
Physical Addressing (MAC & LLC)
bits
physical
Media, Signal and Binary Transmission

OSI Layer Explanation

<table>
<tr><th>OSI Model</th><th>DoD Model</th><th colspan="2">protocols</th><th>devices/apps</th></tr>
<tr><td>layer 5, 6, 7</td><td>application</td><td colspan="2">dns, dhcp, ntp, snmp, https, ftp, ssh, telnet, http, pop3... others</td><td>web server, mail server, browser, mail client...</td></tr>
<tr><td>layer 4</td><td>host-to-host</td><td>tcp</td><td>udp</td><td>gateway</td></tr>
<tr><td>layer 3</td><td>internet</td><td colspan="2">ip, icmp, igmp</td><td>router, firewall
layer 3 switch</td></tr>
<tr><td>layer 2</td><td rowspan="2">network access</td><td colspan="2">arp (mac), rarp</td><td>bridge
layer 2 switch</td></tr>
<tr><td>layer 1</td><td colspan="2">ethernet, token ring</td><td>hub</td></tr>
</table>

Summary of Layers



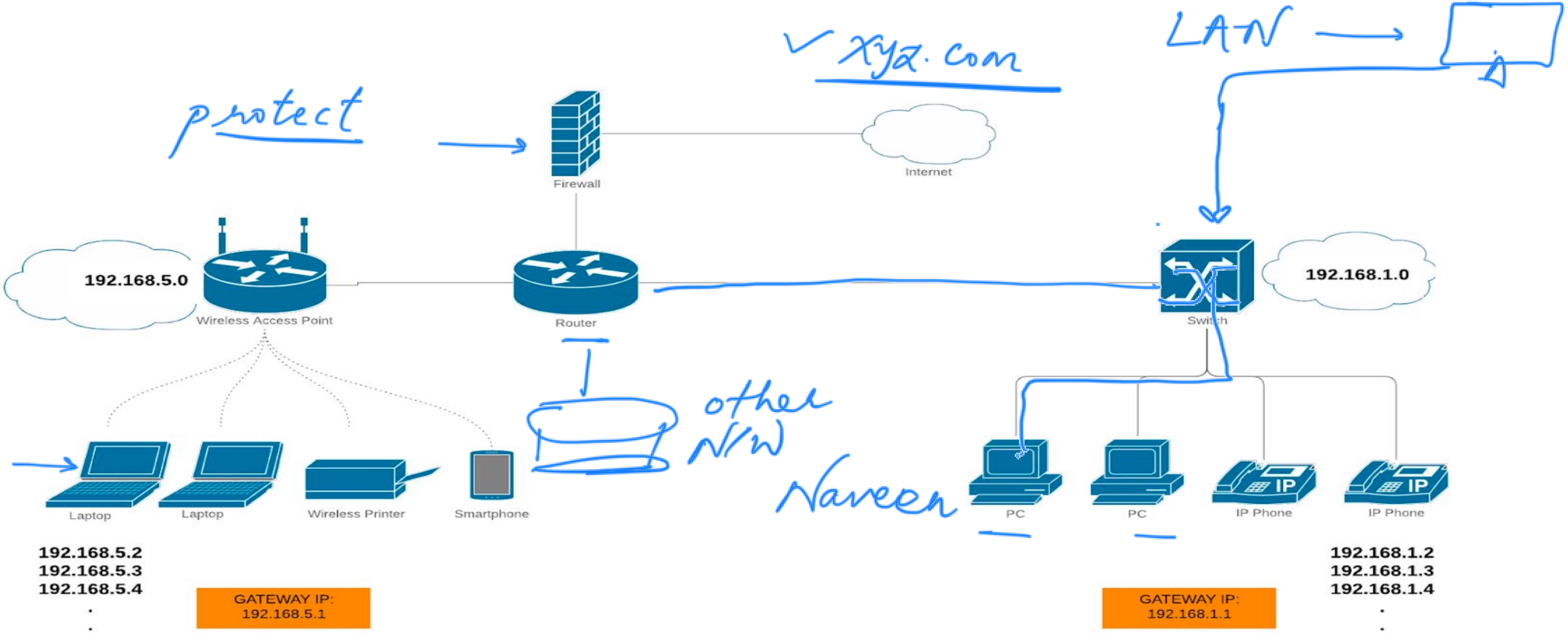
Classification of Network by Geography

- LAN -> Local Area network
- WAN -> Wide Area Network
- MAN -> Metropolitan Area Network
- CAN -> Campus Area Network
- PAN -> Personal Area Network

Networking Pre-requisites

- Switching
- Routing
- Default Gateway
- DNS configuration

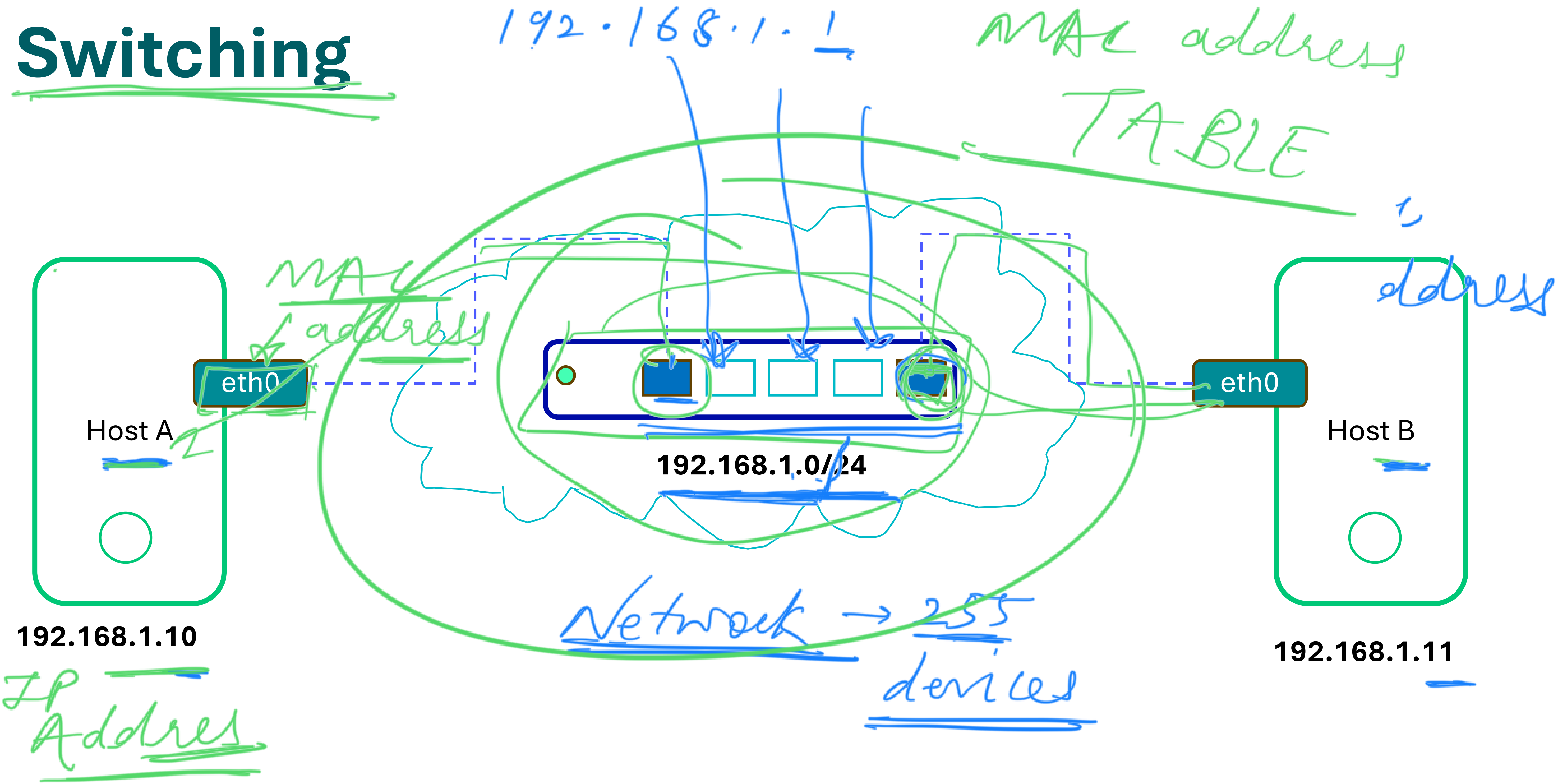
General Networking Diagram



Switches

Switches facilitate the sharing of resources by connecting together all the devices, including computers, printers, and servers, in a small business network

Switching


192.168.1.1
MAC address
TABLE
MAC address
address
eth0
Host A
192.168.1.0/24
eth0
Host B
192.168.1.10
192.168.1.11
IP Address
Network → 255 devices

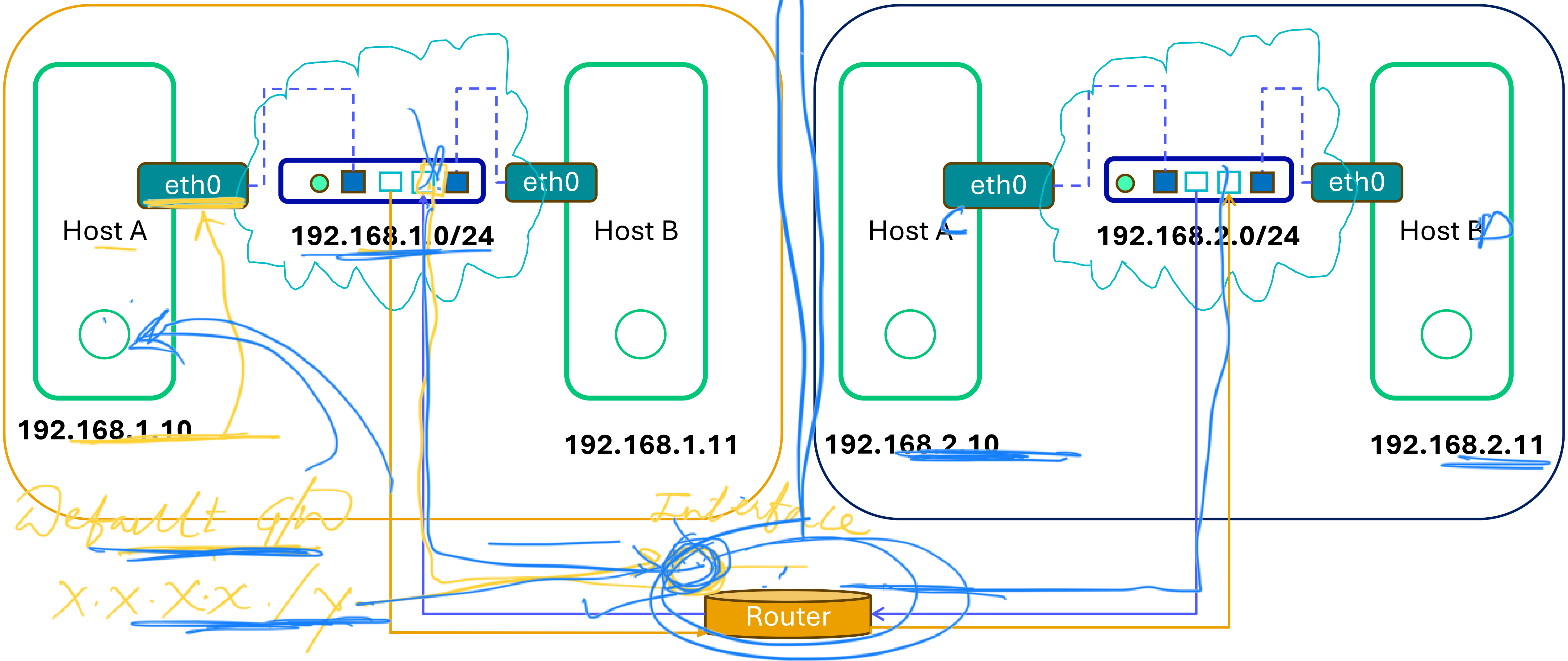
Routers

A router receives and sends data on computer networks. Routers are sometimes confused with network hubs, modems, or network switches. However, routers can combine Multiple Networks together.

Home Network

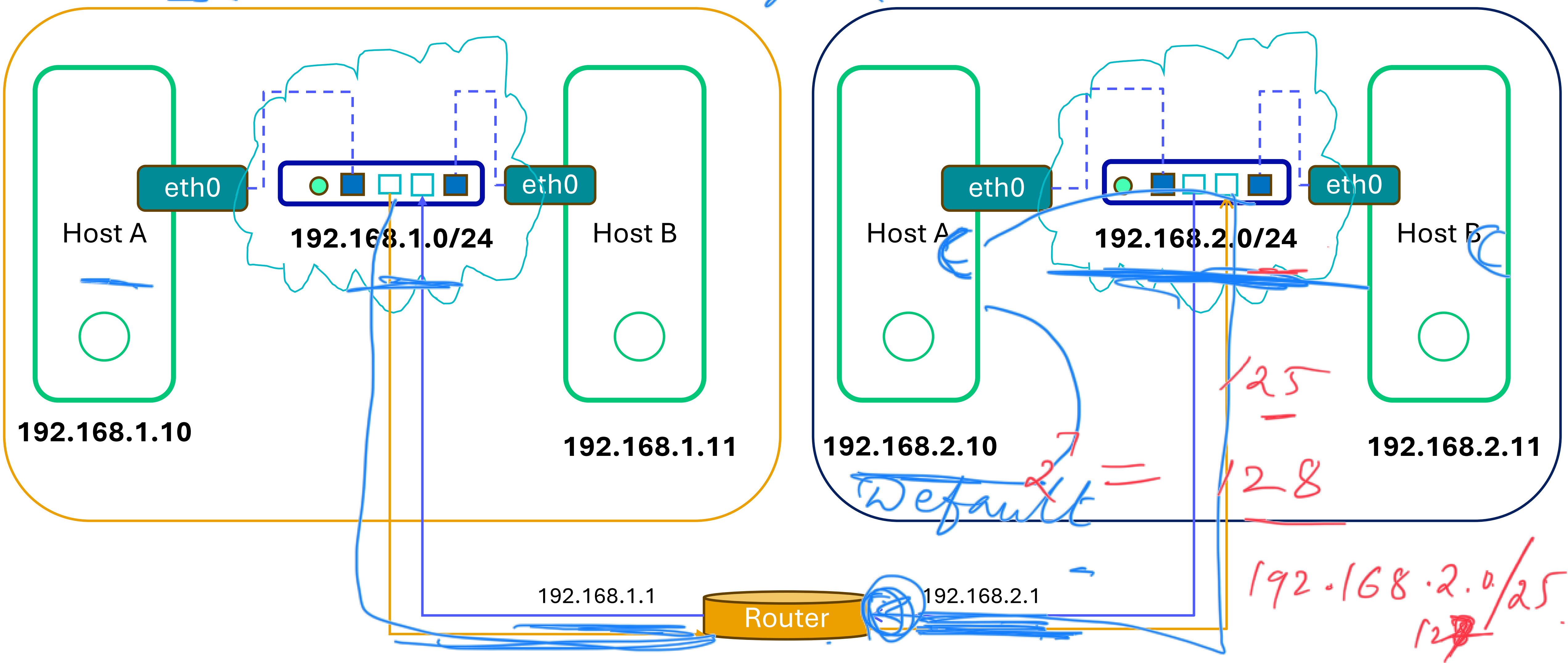
Routing

google.com IP add

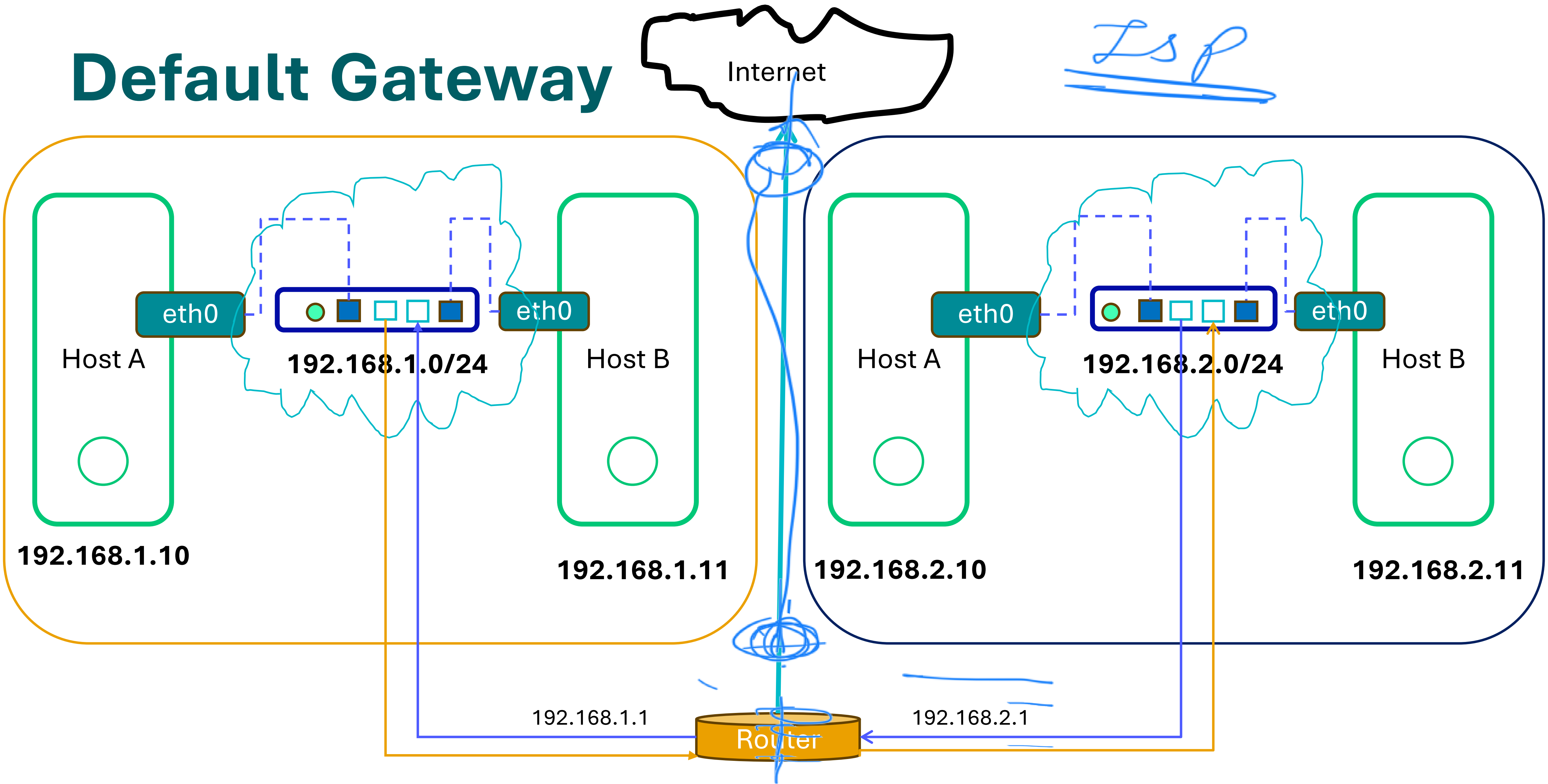


Default Gateway

Google



Default Gateway

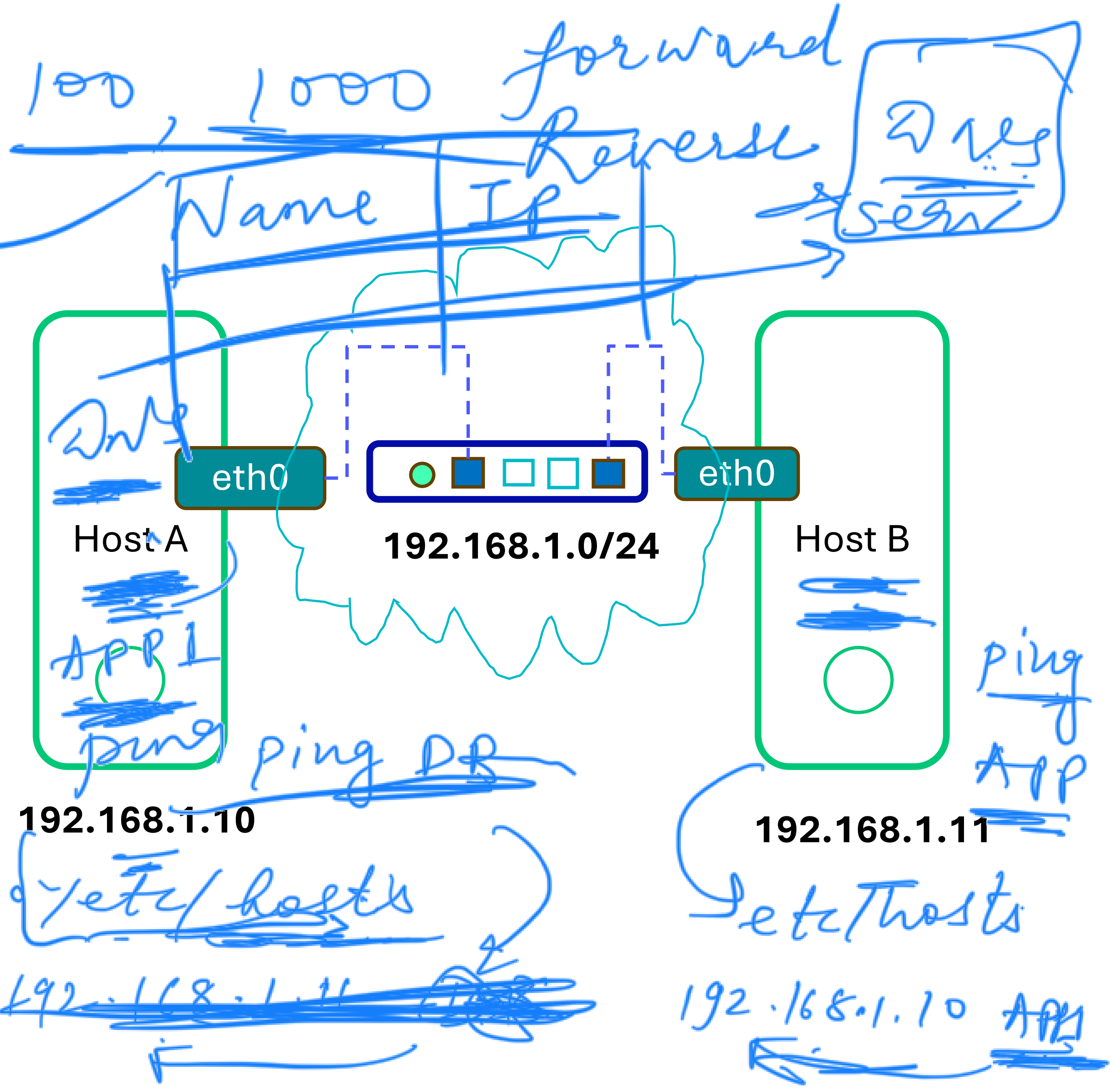


DNS Configurations

cat /etc/hosts

```
atulm@Atuls-MacBook-Air ~ % cat /etc/hosts
##
# Host Database
#
# localhost is used to configure the loopback interface
# when the system is booting.  Do not change this entry.
##
127.0.0.1       localhost
255.255.255.255 broadcasthost
::1             localhost
atulm@Atuls-MacBook-Air ~ %
```

```
atulm@Atuls-MacBook-Air ~ % ping localhost
PING localhost (127.0.0.1): 56 data bytes
Request timeout for icmp_seq 0
Request timeout for icmp_seq 1
Request timeout for icmp_seq 2
Request timeout for icmp_seq 3
^C
```



How DNS Works?

User enters URL in browser (e.g., www.google.com)

Browser checks its own DNS cache

Operating System DNS cache lookup

Windows:
`C:\Windows\System2\drivers\etc\hosts
Linux/Mac: /etc/hosts

Recursive Resolver queries ROOT DNS servers

Recursive Resolver checks its own DNS cache

Send DNS query to configured DNS Resolver
(Your router, corporate DNS, ISP DNS, or public DNS like 8.8.8.8)

Recursive Resolver queries ROOT DNS servers

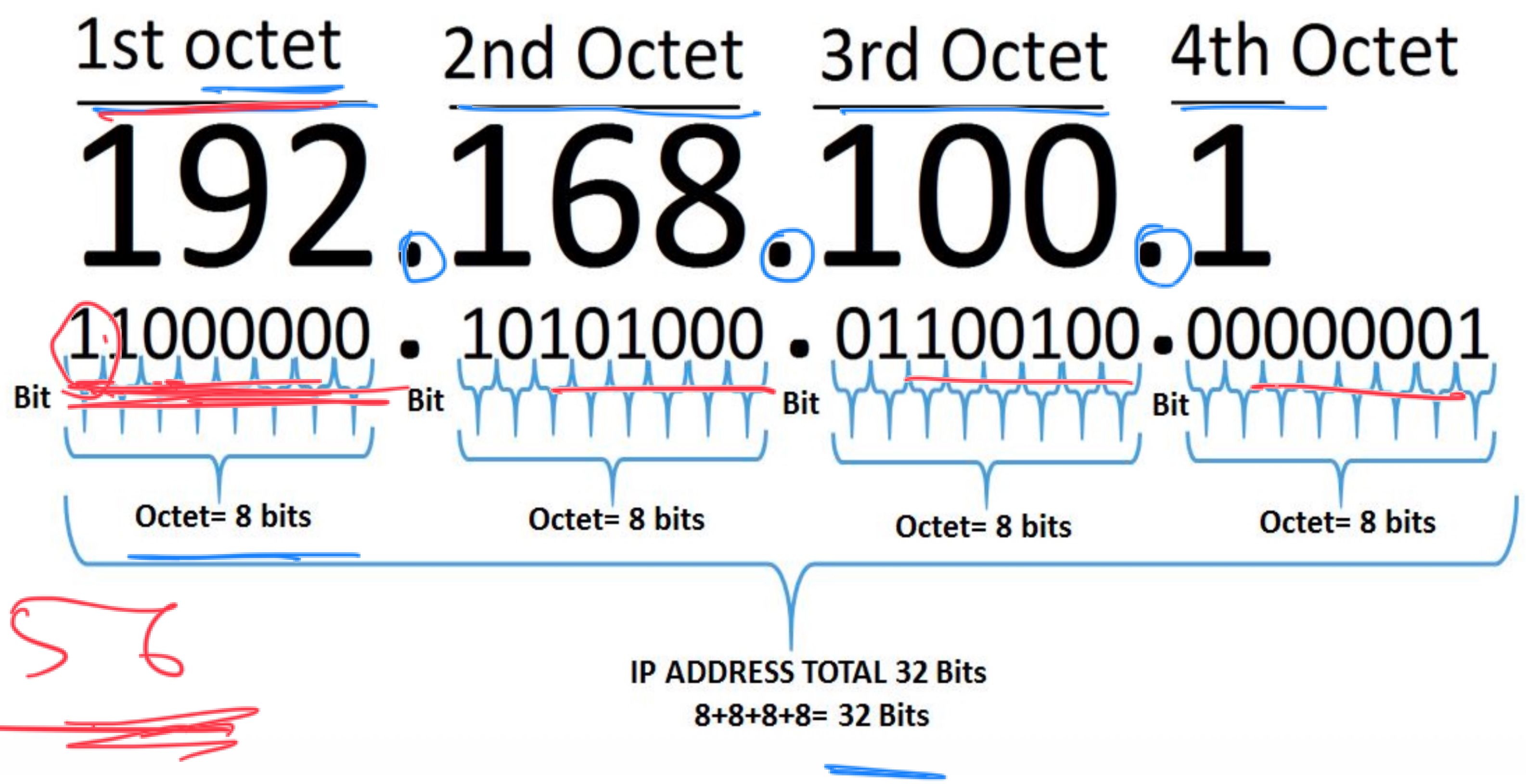
Root servers respond with .COM TLD name servers

Recursive Resolver queries .COM TLD servers

Google's

IPv4 Address

192.168.1.1/32 → 1 IP address



0
1

$2^8 = 256$

IPv4 Range

- 0.0.0.0 – 255.255.255.255
- 00000000.00000000.00000000.00000000 (0.0.0.0)
- 11111111.11111111.11111111.11111111 (255.255.255.255)

Public and Private IP Division

- Public IP => Internet
  - E:g 54.86.23.90
- Private IP => For local network design
  - E:g 192.168.1.10

Private IP Ranges

- Class A 10.0.0.0 - 10.255.255.255
- Class B 172.16.0.0 - 172.31.255.255
- Class C 192.168.0.0 - 192.168.255.255

- Subnet Masks - Example
  - 255.0.0.0
  - 255.255.0.0
  - 255.255.255.0

Protocols

Label on Column	Service Name	UDP and TCP Port Numbers Included
DNS	Domain Name Service – UDP	UDP 53
DNS TCP	Domain Name Service – TCP	TCP 53
HTTP	Web	TCP 80
HTTPS	Secure Web (SSL)	TCP 443
SMTP	Simple Mail Transport	TCP 25
POP	Post Office Protocol	TCP 109, 110
SNMP	Simple Network Management	TCP 161,162 UDP 161,162
TELNET	Telnet Terminal	TCP 23
FTP	File Transfer Protocol	TCP 20,21
SSH	Secure Shell (terminal)	TCP 22
AFP IP	Apple File Protocol/IP	TCP 447, 548

- In the networking and communications area, a protocol is the formal specification that defines the procedures that must be followed when transmitting or receiving data. Protocols define the format, timing,
- sequence, and error checking used on the network.